HERISLEI COELHO PIMENTEL

PhD Applicant - Fall 2027 | Trustworthy AI | Reliability Engineering | Decision Intelligence | Railway Systems

Ipatinga, Minas Gerais, Brazil | herislei@gmail.com | LinkedIn | Google Scholar | ORCID | Research Portfolio

RESEARCH PROFILE

Electrical Engineer and Information Systems graduate with 27+ years of railway and industrial experience in reliability engineering, predictive maintenance, asset management, operations, automation, and safety-critical decision-making. Research interests focus on trustworthy and explainable AI, reliability and degradation modeling, optimization, human-in-the-loop decision support, digital twins, and critical infrastructure. Current work translates real engineering problems into reproducible, data-driven studies with explicit validation, sensitivity analysis, failure analysis, and practical decision value. In 2026, five CNEG & INOVARSE papers were formally certified as published and presented.

RESEARCH INTERESTS

Trustworthy & Explainable AI | Reliability & Degradation Modeling | Optimization | Human-in-the-Loop Decision Support | Digital Twins | Railway Asset Management | Critical Infrastructure

EDUCATION

B.S. Electrical Engineering - Centro Universitário Internacional UNINTER, Brazil, 2022 | GPA 3.91/4.00
B.S. Information Systems - Faculdade Pereira de Freitas / degree registered by UFLA, Brazil, 2009 | GPA 3.78/4.00
Post-Graduate Specialization - Railway Engineering - UniBF, Brazil, 2021 | GPA 4.00/4.00
Post-Graduate Specialization - Maintenance Engineering & Management - UniBF, Brazil, 2021 | GPA 4.00/4.00
Post-Graduate Executive MBA - Metro & Railway Transportation Management - UniBF, Brazil, 2021 | GPA 4.00/4.00
Post-Graduate MBA - Project Management - FGV, Brazil, 2012 | GPA 3.56/4.00
Post-Graduate MBA - Strategic Planning & Management - FACINTER, Brazil, 2010 | GPA 3.52/4.00
Note: Brazilian lato sensu postgraduate specializations/MBAs are listed as postgraduate credentials and are not represented as U.S.-equivalent master's degrees.

PUBLIC PREPRINT, CONFERENCE PUBLICATIONS & SELECTED RESEARCH OUTPUTS

- **Deployment-Aligned Evaluation with Temporal and Geographic Holdouts: A Cross-Case Empirical Study of Forecasting and Semantic Segmentation.** SSRN Preprint, 2026. DOI: 10.2139/ssrn.7538179. Cross-case evaluation of temporal and geographic holdouts using Capital Bikeshare forecasting and Inria building-footprint segmentation; emphasizes leakage control, reproducibility, failure analysis, and bounded generalization claims. Public reproducibility repository.
- **Explainable Maintenance Demand Forecasting for Industrial Railway Locomotives Using Historical SAP Work-Order Data.** Conference publication and presentation certified, XX CNEG & XII INOVARSE 2026; public DOI registration pending. Analyzed 16,490 SAP PM work orders; 13,120 valid records retained. Poisson XGBoost achieved MAE 1.73 and RMSE 2.20, reducing MAE by 12.0% versus a 3-month moving average and 31.2% versus the previous-month baseline; SHAP used for interpretation.
- **Explainable Prioritization of Industrial Railway Wagons: Temporal Validation Using SAP Maintenance Data.** Conference publication and presentation certified, XX CNEG & XII INOVARSE 2026; public DOI registration pending. Developed a Maintenance Attention Index combining recent workload, recency, text-family diversity, and ABC criticality; out-of-sample validation showed positive association with future maintenance demand.
- **ENG-Copilot: Human-in-the-Loop Generative AI for Engineering Decisions and Sustainable People and Process Management.** Conference publication and presentation certified, XX CNEG & XII INOVARSE 2026; public DOI registration pending. Design Science Research framework for traceable engineering decision support with a human authority gate, evidence sufficiency, resource/competency readiness, and sustainability criteria.
- **Railway Benchmarking with Public Data: A Transparent Utilization-Intensity and Safety Index for Brazilian Rail Concessions.** Conference publication and presentation certified, XX CNEG & XII INOVARSE 2026; public DOI registration pending. Reproducible benchmarking across 13 concessions; sensitivity analysis produced Spearman correlation of 0.994 with the baseline ordering.
- **Text Mining of SAP Work Orders to Identify Recurring Patterns and Standardization Opportunities in Industrial Mobile Equipment Maintenance.** Conference publication and presentation certified, XX CNEG & XII INOVARSE 2026; public DOI registration pending. Applies maintenance-text mining to identify recurring themes and standardization opportunities in SAP PM work-order histories.
- **Intelligent Railway Capacity Management in Brazil: MIGIC-Ferro 2.0.** Submitted manuscript, BNDES Railways Award 2026. Data-driven public-policy architecture integrating governance, operational benchmarking, institutional maturity, and progressive Digital Twin simulation across 13 Brazilian rail concessions.
- **Energy for Entrepreneurship: Electricity Reliability, Small-Business Entry, and Territorial Prioritization through IVEP-PAEER in Brazil (2017-2025).** Award-submission manuscript, Sebrae Small Business Economics Award 2026. Econometric panel of 599,359 municipality-month observations covering 5,567 municipalities.

COMPUTATIONAL & RESEARCH METHODS

- Python, R, Power BI, SAP PM/MM/PP; applied machine learning, statistical modeling, econometrics, reliability/failure data, maintenance histories, and engineering dashboards.
- Machine learning: XGBoost, Histogram Gradient Boosting, Random Forest, U-Net/ResNet-34, SHAP, temporal validation, walk-forward evaluation, geographic holdout, data-leakage control, sensitivity analysis, and failure analysis.
- Reliability & asset management: MTBF, MTTR, RCA, FMEA, RCM, predictive maintenance, vibration analysis, ultrasonic inspection, criticality, and lifecycle decision support.
- Research methods: fixed-effects panel models, Negative Binomial regression, Design Science Research, composite indices, reproducibility, out-of-sample validation, and interpretable decision support.

PROFESSIONAL EXPERIENCE

Railway Maintenance Coordinator / Maintenance Coordinator II - Usiminas, Brazil | Jan 2016-Present

- Lead maintenance, reliability, inspection, engineering improvement, and asset-management strategies for critical railway and industrial assets in continuous operation.
- Coordinate approximately 100 direct and contractor professionals across multidisciplinary maintenance activities, integrating Maintenance, Operations, Engineering, Planning, Supply, Safety, and vendors.
- Asset scope includes approximately 32 locomotives, 440 freight railcars, 120 km of railway infrastructure, signaling, turnouts, special machines, and mobile/heavy equipment.
- Use failure history, condition evidence, MTBF, MTTR, availability, asset criticality, SAP PM/MM/PP records, and operational constraints to support intervention, resource, and planning decisions.

Earlier Railway / Maintenance Leadership - Usiminas, Brazil | 1998-2015

- Progressive technical and leadership experience spanning electromechanical maintenance, railway operations, locomotive maintenance, and railway maintenance supervision.
- Led or supported predictive-maintenance, life-extension, failure-elimination, logistics, and reliability initiatives in a large integrated steelmaking environment.

SELECTED ENGINEERING IMPACT

- Reliability initiatives contributed to approximately 30% higher locomotive availability, 50% lower MTTR, and 15% lower annual maintenance costs.
- Vibration-based predictive maintenance increased critical electric drive-motor service life by approximately 300% while reducing maintenance interventions by approximately 85%.
- Ultrasonic condition monitoring extended critical rolling-component service life by approximately 40%.
- Operational and logistics improvements reduced freight-wagon detention by approximately 80% and related operating costs by approximately 60%.

ADDITIONAL ACADEMIC PREPARATION

- HarvardX coursework (audit, 2026): Machine Learning and AI with Python; Using Python for Research; Data Science: Inference and Modeling; Data Science: Building Machine Learning Models; CS50's Introduction to AI with Python.
- Academic foundations in probability/statistics, control, signals and systems, instrumentation, automation, numerical methods, algorithms, databases, operations research, and information systems.

RESEARCH AGENDA

- Trustworthy and explainable AI for reliability-centered maintenance and critical-asset decision support.
- Reliability, degradation, and uncertainty-aware optimization for railway and infrastructure assets.
- Human-in-the-loop decision intelligence and progressive Digital Twin architectures that combine data, simulation, engineering judgment, and operational constraints.
- Failure-aware perception for railway and infrastructure monitoring under distribution shift, uncertainty, and out-of-distribution conditions.

ENGLISH PROFICIENCY & LANGUAGES

- **EF SET English Certificate - 62/100, CEFR C1 Advanced (October 2026)** | Verified Credential.
- Portuguese: Native/Fluent.